

August 5, 2026

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer position, requisition JR100829, with Viridien in Calgary. With a Master of Science in Computer Science from the University of Calgary and applied experience in Python, C++, PyTorch, machine learning, data engineering, and Earth observation data systems, I am excited by the opportunity to contribute to AI tools at a company focused on Earth science, data science, and sustainable technology.

My graduate research focused on developing scalable software and machine learning workflows for large geospatial and Earth observation datasets. As a research software developer with BigGeo, Mitacs, and the University of Calgary, I built C++ and Python systems for processing, encoding, storing, and retrieving satellite imagery. I developed DGGs-based pipelines that converted imagery into structured tensor files, improved processing performance with C++ multithreading, and worked with GDAL, Rasterio, SNAP, and satellite data APIs. This background maps closely to Viridien's need for engineers who can combine data structuring, scientific computing, software engineering, and applied ML thinking.

I also bring hands-on deep learning experience from my digital elevation model super-resolution project. I built Python and PyTorch workflows for data collection, preprocessing, model training, and evaluation, trained deep ResNet models for 4x terrain reconstruction, and assessed model performance using RMSE, slope, TRI, TPI, and wavelet-based metrics. Through this work, I became comfortable running experiments, comparing baselines, diagnosing model limitations, and communicating results in a way that supports better technical decisions.

Beyond research, I have backend engineering experience developing Django REST APIs backed by PostgreSQL, and I have supported students with AWS-based data pipelines, large-scale data workflows, SQL, debugging, and project work as a teaching assistant. I am strongest in Python, C++, PyTorch, data pipelines, scientific visualization, and reproducible engineering. I am also actively expanding my LLM and AI-agent workflow experience through my own job-search automation platform, where I use LLM-based matching, workflow automation, Google Sheets, Google Drive, Docker, and agent-style orchestration.

Viridien interests me because the role sits directly at the intersection of machine learning, Earth data, visualization, experimentation, and practical AI systems. I would be excited to bring my Calgary-based research background, programming discipline, data-engineering experience, and curiosity for scientific problem solving to your team.

Thank you for considering my application. I would welcome the opportunity to discuss how my machine learning, Earth observation, and software engineering experience can contribute to Viridien's AI and data science work.

Sincerely,
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